Problem 6

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Problem 1 *Estimate the amount of paint needed to apply a coat of paint .05 cm thick to a hemispherical dome with diameter 50m. Is this value an underestimate or an overestimate?*

Explanation. The radius of the dome is $\frac{50}{2}m = 25m$. The paint increases this by 0.0005m (0.05cm to meters). The volume of a "hemispherical dome" (or half of a sphere) is $V = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi r^3$. Then

$$dV = 2\pi r^2 \, dr$$

The amount of paint needed is approximately the change in volume (dV). We also have that $r = 25m$ and $dr = (5 \times 10^{-4})m$. Thus, the amount of paint needed to paint the dome is approximately

$$\Big[dV\Big]_{r=25,\, dr=0.0005} = 2\pi(25)^2(5 \times 10^{-4}) = 5\pi(0.125) = 0.625\pi \approx 1.9635m^3$$

The function $V(r) = \frac{2}{3}\pi r^3$ has derivatives $\frac{dV}{dr} = 2\pi r^2$, and $\frac{d^2V}{dr^2} = 4\pi r$. This second derivative is positive for r in the interval $(25, 25.0005)$, so the graph of V is concave up in that entire interval. This means the estimate is an underestimate.